MILITARY SPECIFICATION

ADDITIVE, ANTIFREEZE EXTENDER, LIQUID COOLING SYSTEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and one grade of additive intended for inhibiting water or reinhibiting used MIL-A-46153 Antifreeze (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

O-I-490	- Inhibitor, Corrosion, Liquid Cooling System.
NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial).
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-C-569	- Container, Plastic, Molded (for Liquids, Pastes and Powders), Overpacked.
PPP-F-320	- Fiberboard; Corrugated and Solid Sheet Stock (Container Grade), and Cut Shapes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6850

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MIL-B-26701	- Bottles, Screwcap and Carboys, Polyetheylene Plastic.
MIL-T-36812	- Test Kit, Reserve Alkalinity, Antifreeze.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

TECHNICAL BULLETIN

TB 750-651	- Use of Antifreeze Solutions and Cleaning Compounds in Engine Cooling Systems

(Copies of specifications, standards and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1121	- Reserve Alkalinity of Engine Antifreeze, and Coolants.
D1122	- Specific Gravity of Engine Antifreezes by the Hydrometer.

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D1287 - pH of Engine Antifreezes, Antirusts and Coolants.

D3634 - Trace Chloride Ion in Engine Antifreezes/ Coolants in the Presence of Mercaptobenzothiazole.

Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania (19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials The raw materials used in the manufacture of the additive shall be intimately mixed and processed so as to produce a clear, uniform solution conforming to all the requirements of this specification.

3.2 Composition. The additive shall consist of an aqueous solution of sodium metaborate, sodium mercaptobenzothiazole, and potassium silicate in the proportions specified in table 1.

TABLE I. Chemical composition.

Components	Weight Percent
Distilled Water (total added)	63.4 ± 0.60
Sodium Metaborate ($Na_2B_2O_4 \cdot 8H_2O$)	29.0 ± 0.30
Potassium Silicate Solution	4.6 ± 0.20
Sodium Mercaptobenzothiazole 50% aqueous solution, by weight	3.0 ± 0.15

3.2.1. Sodium Metaborate. The sodium metaborate ($Na_2B_2O_4 \cdot 8H_2O$) shall be technical grade with a minimum purity of 98 percent which will meet all requirements of this specification.

3.2.2 Potassium silicate The potassium silicate is an aqueous solution containing 38.80 weight percent potassium silicate. The weight ratio of SiO_2 to k_2O shall be 2.1 to 2.10. KASIL #6 manufactured by Philadelphia Quartz Co., P. O. Box 840, Valley Forge, PA 19482 is known to have this composition.

3.2.3 Sodium Mercaptobenzothiazole. The sodium salt of mercaptobenzothiazole shall be a 50 percent agueous solution by weight technical grade which will meet all requirements of this specification.

3.3 Other chemical requirements. In addition to the requirements of 3.2, 3.2.1, 3.2.2, 3.2.3 and table I, the additive shall conform to the chemical requirements specified in 3.3.1 to 3.3.4 inclusive.

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3.3.1 pH value. When tested as specified in 4.5.2, the packaged pH value shall be 12.5 ± 0.5.

3.3.2 Reserve alkalinity. The reserve alkalinity shall be not less than 11.0 when tested as specified in 4.5.3.

3.3.3 Chlorides. The chloride content shall not be more than 10 ppm when tested as specified in 4.5.4.

3.3.4 Insoluble matter. The insoluble matter shall not be greater than 0.02 percent by weight when tested as specified in 4.5.5.

3.4 Physical requirements.

3.4.1 Specific gravity. The specific gravity of the undiluted material shall be no less than 1.185 nor greater than 1.200 at 60°/60° F when determined as specified in 4.5.6.

3.5 Storage stability. Stability shall be such that the additive will conform to this specification after warehouse storage in the original containers for a 12 month period at temperatures of 10° C (50° F) to 43° C (110° F).

3.6 Workmanship. The additive shall be processed to produce a clear, uniform product which complies with the requirements of this specification.

4. QUALITY ASSURANCE PROVISION.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. All compounded material of the same type manufactured as one batch and offered for delivery at one time shall be considered a lot or purposes of acceptance inspection and tests.

4.3.1 Sampling for inspection. Sampling for inspection and tests shall be performed in accordance with MIL-STD-105.

4.3.2 Examination of the antifreeze extender additive. Visual examination of the antifreeze extender additive shall be made to determine compliance with specification requirements on presence of foreign particulate matter, and single-phase homogeneity. The sample unit shall be one quart container. The inspection level shall be S-3 and acceptable quality level of 2.5 percent defective for major defects.

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4.3.3 Examination of the antifreeze extender additive for defects in net contents. The sample unit shall be one filled quart container. The inspection level shall be S-2 or a minimum of 5 sample units randomly selected from the lot. The average net contents shall not be less than the amount specified.

4.3.4 Quality conformance testing of the antifreeze extender additive. The end item shall be tested for conformance to the requirements as specified in Section 3 for each lot presented for inspection. The lot size shall be expressed in units of quarts of additive. The sample unit shall be four one-quart containers. No more than one sample unit shall be taken from any one shipping container. The inspection level shall be S-1 with no less than three sample units selected. There shall be no evidence of failure to meet the specified requirements.

4.4 Lot acceptance tests. The samples shall be subjected separately to the tests specified. If either specimen fails one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests, provided the contractor has removed or reworked all nonconforming material.

4.5 Tests.

4.5.1 Chemical composition. Any quantitative spectrographic or wet chemical method that can be shown to produce the accuracy necessary to assure conformance of the material to the requirements in table I may be used. Certification of the test method is required prior to acceptance by the Government. Application for certification should be forwarded to U.S. Army Mobility Equipment Research and Development Command, ATTN: DRDME-GL, Ft. Belvoir, Va. 22060.

4.5.2. pH value. The pH determination of the packaged material shall be made at 25 $\pm$ 2° C, using the apparatus and procedure specified in ASTM Method D1287.

4.5.3 Reserve alkalinity. The reserve alkalinity shall be determined in accordance with ASTM Method D1121.

4.5.4 Chlorides. The chloride content shall be determined in accordance with ASTM Method D3634.

4.5.5 Insoluble matter. Weigh 500.0 + 0.1 grams of the additive from a well agitated unopened container into a 600 ml beaker. Filter through a medium porosity tared Gooch filtering crucible, rinse the beaker thoroughly with distilled water, completely transfering all insoluble matter to the crucible. Wash five times with 15 ml portions of distilled water and dry to constant weight at 105° C. Gain in weight of the crucible is insoluble matter.

4.5.6 Specific gravity. Specific gravity of the packaged material shall be determined in accordance with ASTM Method D1122.

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4.6 Storage stability In lieu of testing, the manufacturer shall certify that the formula supplied under this specification has been stored in the plastic containers specified in 5.1 and found to meet the requirements of the specification after storage. The containers shall not show disintegration due to the action of the additive.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit product.

4.7.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.1.3 Examination. Samples selected in accordance with 4.7.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101.	Materials and containers not as specifed for level A or B. Each incorrect material or container shall be considered one defect.
102.	Quantities packed together not as specified for level A or B.
103.	Box not provided with a liner, pads and partitions as specified level A or B.
104.	Preservation and packing not in accordance with the referenced document as specified for commercial.
105.	Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial (see 6.2).

5.1.1 Level A. The additive shall be unit packed in a one quart capacity container. The container shall conform to PPP-C-569, Type 1, Class A or MIL-B-26701. When a bottle conforming to MIL-B-26701 is used, the bottle shall be manufactured using an additive in the polyethylene resin which will yield a bottle that no portion will transmit more than 1 percent of ultraviolet light at any wavelength in the range of 310 to 325 millimicrons when tested using a calibrated spectrophotometer with air as a reference (see 6.3).

5.1.2 Commercial. The additive shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. Twelve one quart bottles of additive shall be packed together, in a single layer, using a close fitting box conforming to PPP-B-601, Overseas Type, style optional or PPP-B-621, Class 2, style optional.

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The box shall be provided with a full liner, top and bottom pads, and interlocking partitions to separate the bottles; fabricated from fiberboard conforming to PPP-F-320, Type CF, Class weather-resistant, variety SW, Grade W5c. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A except the box shall conform to PPP-B-636, V3c.

5.2.3 Commercial. The additive shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

5.4 Palletization. When specified in the contract or order (see 6.2) the packed additive for level A or B shall be palletized in accordance with MIL-STD-147, load type as applicable, using pallets conforming to NN-P-71, Type IV. Commercial packing may utilize a commercial type pallet suitable for the weight and size involved.

6. NOTES

6.1. Intended use. The additive is intended for use in used MIL-A-46153 antifreeze and in water at a concentration of three percent (one-half quart additive per seventeen quart antifreeze or water). Used antifreeze is defined as antifreeze that tests green to yellow when tested as specified in TB-750-651 using MIL-T-36812, Test Kit, Reserve Alkalinity, Antifreeze. Additive is to be used in lieu of O-I-490 in all military vehicles or equipment using MIL-A-46153 antifreeze or water. It is not intended for use in commercial antifreeze.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in acquisition documents:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1.)
c. Degree of preservation and degree of packing required (see 5.1 5.2).
d. When palletization is required (see 5.4).

6.3 Additive to plastic containers. An additive which is known to be effective in excluding ultraviolet light from the interior of polyethylene containers is phthalocyanine blue.

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6.4 Preparation of additive

6.4.1 Mixing Sequence. In order to obtain a clear solution the following mixing sequence should be followed:

a. Weigh distilled water.
b. Heat at 38° C (100° F) and stir in weighed. sodium. metaborate until solution is clear.
c. Cool to room temperature (25° C).
d. Add weighed potassium silicate using the water/borate
e. Add weighed sodium mercaptobenzothiazole and continue stirring until all components are in solution approximately 30 minutes.

Custodians:
Army - ME
Navy - SH
Air Force - 68

Preparing activity:
Army - ME

Project 6850-0684

Review activities:
Army - AT, SM
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User activity:
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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-A-53009 Additive, Antifreeze Extender, Liquid Cooling Systems

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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